

Akanksha Singh

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About Me

Researcher in machine learning, specialising in deep learning, vision, and speech. My broad aim is to advance responsible AI research, enhancing trustworthiness, with a particular focus on fairness, inclusion, explainability, privacy and societal impact.

Education

Indian Institute of Science Education and Research, Bhopal

Jul 2019 - Jul 2024

BS-MS in *Electrical Engineering and Computer Science* | CPI : 8.36/10

- Minor: Data Science and Engineering
- Relevant Courses: Calculus, Computer Vision, Deep Learning, Linear Algebra, Machine Learning, Natural Language Processing, Probability, Statistics

Experience

Centre for Responsible AI (CeRAI), IIT Madras

Aug 2024 - Mar 2026

Post-Baccalaureate Fellow | Supervisor: *Prof. Balaraman Ravindran*

- Research on failure modes - high false positive rate, real-class collapse, dataset and training bias, poor generalisation - of audio deepfake detection and proposal of deployment-ready evaluation framework based on cost-based selective classification.
- At the discussion stage of the project on multi-agent systems for multimodal deepfake detection, in tandem with the RAG framework to provide additional support through fact-checking.
- Research on explainable audio deepfake detector using linguistic-motivated artefacts.
- Chapter author of the "Grounding Responsible AI in Agriculture", a project in collaboration with Meta OpenLoop. The work explores current applications and associated risks of AI in agriculture, identifies key gaps in its development and deployment, and examines relevant policies, regulations, and legal frameworks. The paper emphasises the importance of responsible AI through actionable technical and policy recommendations, grounded in expert interviews and comprehensive analysis research.

Image Analysis and Biometrics Lab, Department of CSE, IIT Jodhpur

May 2023 - Jul 2024

External Master's Thesis | Supervisor: *Prof. Mayank Vatsa & Prof. Richa Singh*

- Created and curated a large (approx 1.3M samples), diverse (26 generation techniques across 4 sets of real-fake content of audio-video modalities), balanced (wrt sex and skin-tone) benchmark multimodal deepfake dataset comprising identity-aware swaps and synthetic media.
- Performed qualitative experiments such as BRISQUE score for visual quality and FAD for audio quality. And benchmarked using 12 existing state-of-the-art unimodal (visual and audio) and multimodal deepfake detection models.

Publications

Conference **ILLUSION: Unveiling Truth with a Comprehensive Multi-Modal, Multi-Lingual Deepfake Dataset**
Kartik Thakral*, Rishabh Ranjan*, **Akanksha Singh**, Akshat Jain, Mayank Vatsa, Richa Singh
International Conference on Learning Representations (ICLR) 2025
* = equal contribution

Skills

Languages Python, C/C++

Frameworks PyTorch, HuggingFace, Dockers, $\LaTeX$, Git, Bash, Slurm

Presentations

Poster Explainable Audio Deepfake Detection, Conclave on AI Governance, CeRAI IIT Madras
Presentation Explainable Multilingual Audio Deepfake Detection, WSAI Bi-weekly, WSAI IIT Madras
Poster ILLUSION, ICLR 2025
Presentation Deepfakes & ILLUSION, Group Weekly, IIT Madras

Achievements

2024 Women Post Baccalaureate Fellow, Research Fellowship, CeRAI, IIT Madras

2023 Qualified for Assistant Professor, Subject: CSA, UCG-NET

Events

<i>School</i>	Attended the Winter School on Foundation Models (FOMO) 2026 organised by ELLIS ELIAS ELLIOT at Tolhuistuin, Amsterdam, in Mar 2026.
<i>Conclave</i>	Volunteered at the Conclave on Safe and Trusted AI, organised by the CeRAI, IIT Madras, in Dec 2025.
<i>Event</i>	Attended the TIACON 2025, an event on Information & Trust in the AI age organised by Trusted Information Alliance at the India Habitat Centre, New Delhi, in Nov 2025.
<i>Conclave</i>	Presented at the Conclave on AI Governance organised by CeRAI, IIT Madras in Oct 2025.
<i>Summit</i>	Attended Microsoft Research India Academic Summit 2025 held at Bengaluru in Jun 2025.
<i>Conference</i>	Attended the ICLR 2025 in Singapore, in Apr 2025
<i>Event</i>	Attended MisinfoCon India, organised by Tattle and Hacks/Hackers at Thoughtworks, Bengaluru, in Mar 2025.
<i>Conference</i>	Attended CODS-COMAD 2025, a data-centric conference in India, held in IIT Jodhpur, in Dec 2024.

Academic Projects

Interpretable Deep Learning for Text Classification

- Implemented an existing novel interpretability framework for text CNNs from scratch to linguistically motivated features. The proposed features are borrowed from existing work on text classification of shorter texts between fiction and non-fiction genres (Brown Corpus).
- The exit criteria are to corroborate our interpretability results with the baseline results obtained from the existing work using logistic regression.

Consumer Complaints: Text Classification Problem

- Used machine learning and deep learning models for a multi-class nominal label classification.
- A vectorisation and feature selection pipeline was used for statistical architecture, and hyperparameters were fed to a cross-validation grid search on 5 supervised learning models. Logistic regression gave the best results, accuracy of 85.9%.
- For DL architecture, FFN, bi-directional LSTM, RNN, and transformers were used. Bi-directional LSTM gave the best results, with an accuracy of 86.5%.

Teaching Experience

ECS308: Data Science and Machine Learning

Jan 2023 - Apr 2023

Undergraduate Teaching Assistant

- Course Instructor: Dr. Tanmay Basu. One of the two TAs for the course with 50 students.
- Conducted labs on supervised and unsupervised machine learning and hyperparameter tuning.
- Graded assignments, course projects, quizzes and examinations.

Personal Interests

<i>Sports</i>	Served as the first female Secretary for the Sports Society. Participated in the Inter IISER Sports Meet in the sports basketball, football, and athletics, and won Silver (2019) and Gold (2022 and 2023) in Basketball. Served as Basketball Coordinator.
<i>Volunteer</i>	Volunteered as a Peer (Student) Counsellor to the Institute Counselling Cell.
<i>Quiz</i>	Served as the Publicity and PR Coordinator for the IBQC-Quizzing Society.
<i>Dance</i>	Served as the Committee Member of the Mayuraa-Dance Society.
<i>Arts</i>	Served as the Vice-Secretary of the FALC-Fine Arts and Literary Society.
<i>Others</i>	Completed two Himalayan treks and ran a half-marathon.